|INCH-POUND|

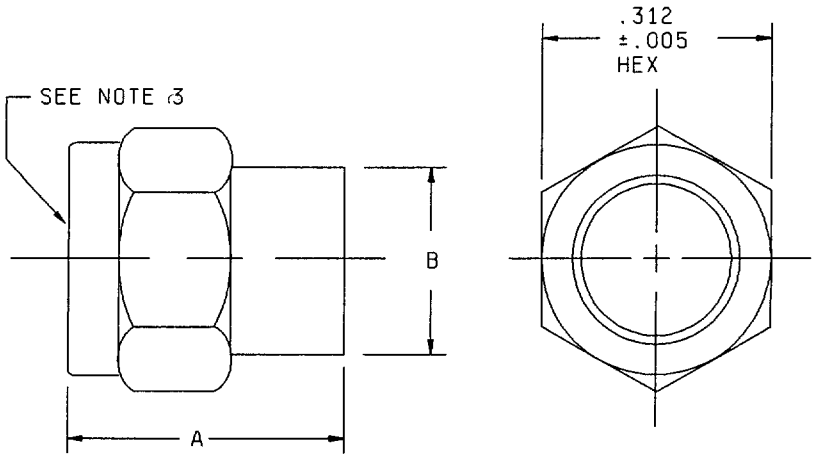
MIL-D-39030/3B
1 March 1994
SUPERSEDING
MIL-D-39030/3A
22 December 1980

MILITARY SPECIFICATION SHEET

DUMMY LOADS, ELECTRICAL, COAXIAL, TYPE I (SMA), LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the dummy loads described herein shall consist of this document and the latest issue of MIL-D-39030.



Dash number	A max	B max
-01, -16	.73	.28
-02, -17	.73	.28
-11, -22	.43	.27
-15, -26	.56	.28

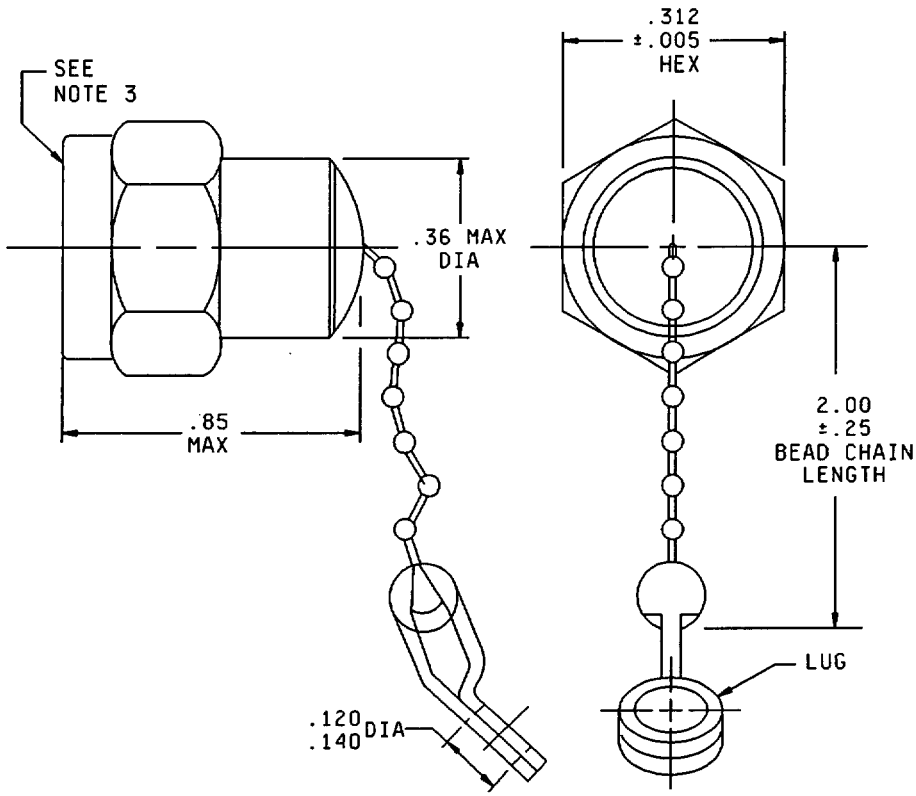
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Series SMA pin contact interface IAW MIL-STD-348.

Inches	mm
.005	0.13
.27	6.9
.28	7.1
.312	7.92
.43	10.9
.56	14.2
.73	18.5

FIGURE 1. Dimensions and configuration, PIN M39030/3-01, -02, -11, -15, -16, -17, -22, and -26 (both N and S).

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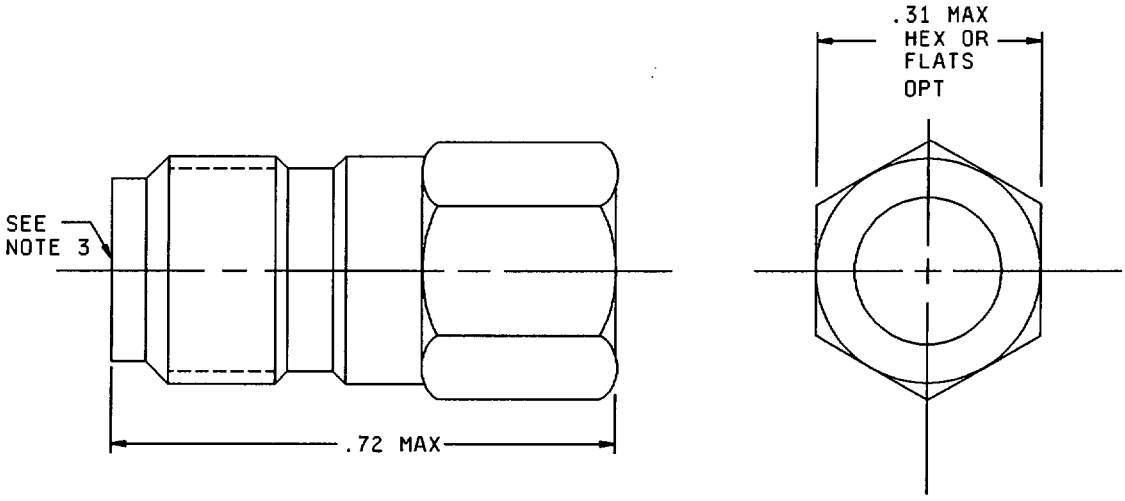


Inches	mm
.005	0.13
.120	3.05
.140	3.56
.25	6.4
.312	7.92
.36	9.1
.85	21.6
2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Series SMA pin contact interface IAW MIL-STD-348.

FIGURE 2. Dimensions and configuration, PIN M39030/3-03, -04, -18, and -19 (both N and S).

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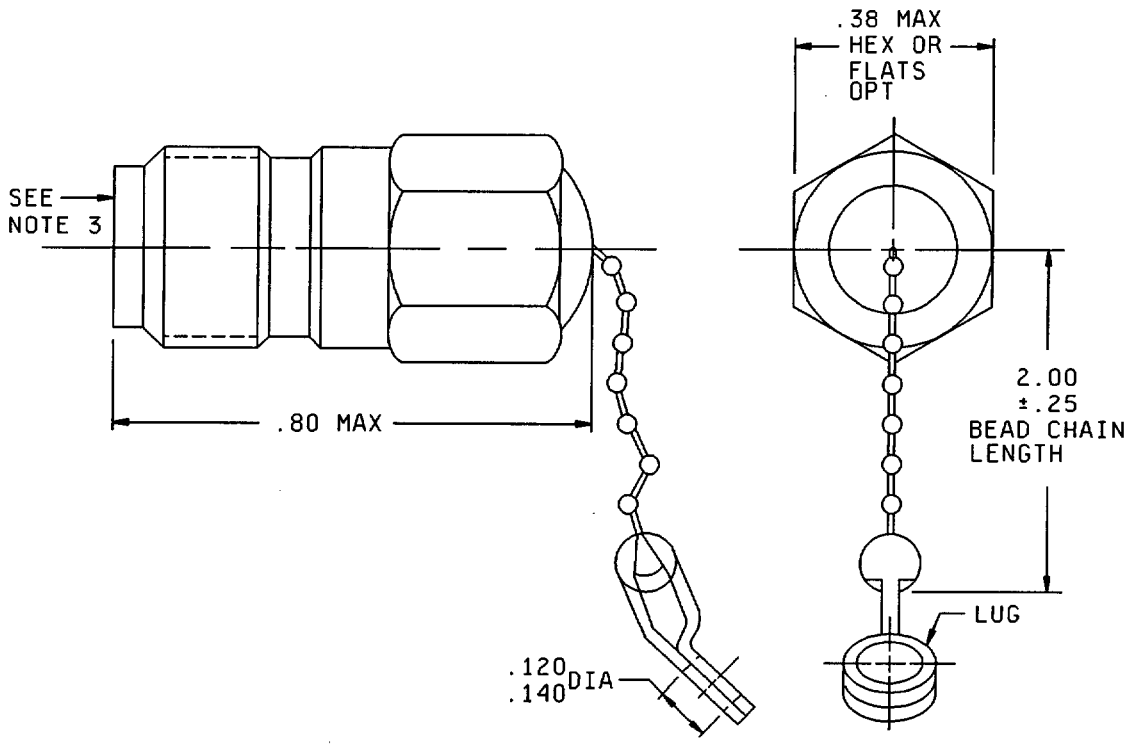


Inches	mm
.315	7.9
.72	18.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Series SMA socket contact interface IAW MIL-STD-348.

FIGURE 3. Dimensions and configuration, PIN M39030/3-05 and -06 (both N and S).

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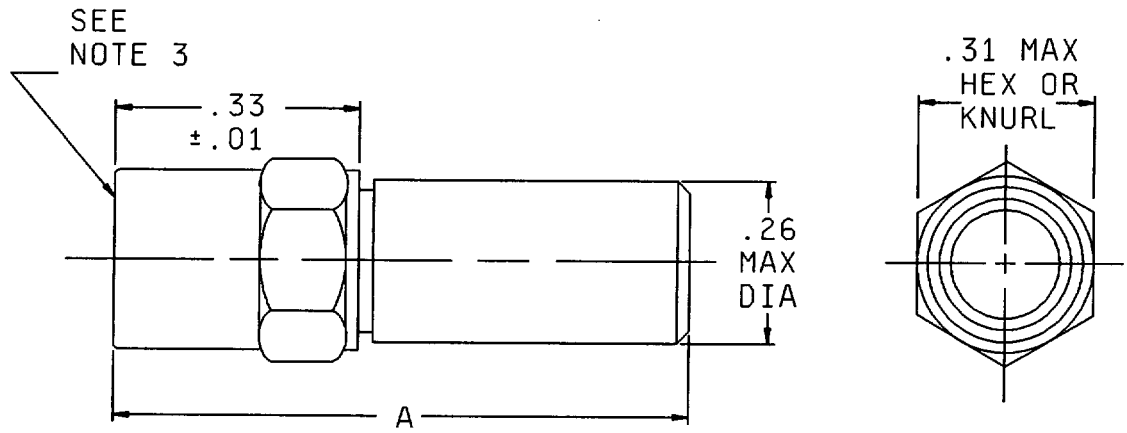


Inches	mm
.120	3.05
.140	3.56
.312	7.92
.38	9.71
.80	20.3
2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Series SMA socket contact interface IAW MIL-STD-348.

FIGURE 4. Dimensions and configuration, PIN M39030/3-07 and -08 (both N and S).

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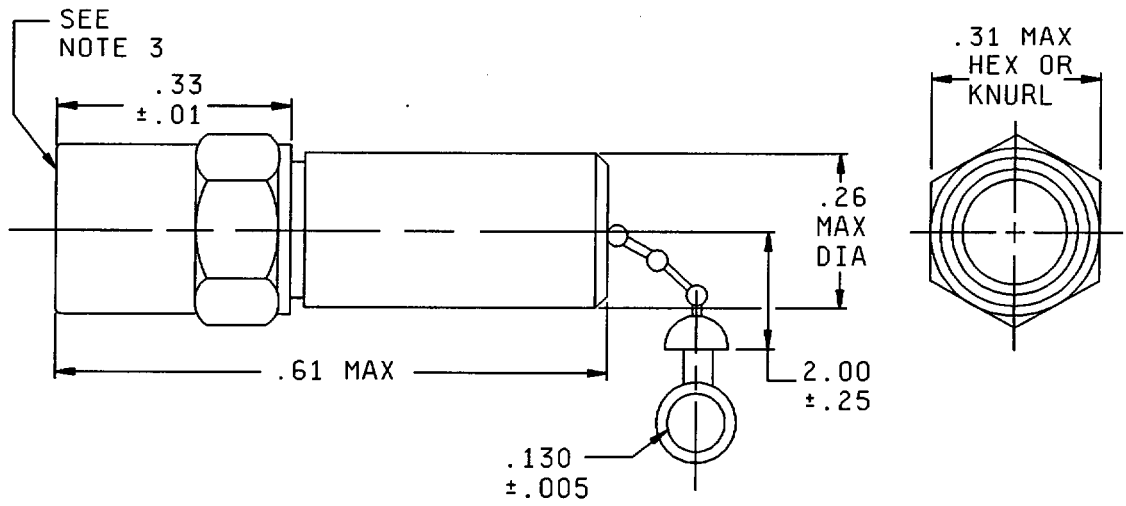
Dash number	A max
-09, -20	.84
-10, -21	.55
-12, -23	.55
-14, -25	.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Series SMA pin contact interface IAW MIL-STD-348.

Inches	mm
.01	0.3
.26	6.6
.31	7.9
.33	8.4
.55	14.0
.84	21.3

FIGURE 5. Dimensions and configuration, PIN M39030/3-09, -10, -12, -14, -20, -21, -23, and -25 (both N and S).

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Inches	mm
.005	0.13
.01	0.3
.130	3.30
.25	6.4
.26	6.6
.31	7.9
.33	8.4
.61	15.5
2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Series SMA pin contact interface IAW MIL-STD-348.

FIGURE 6. Dimensions and configuration, PIN M39030/3-13 and -24 (both N and S).

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TABLE I. Dash numbers and characteristics.

Dash number both N and S	Operating frequency	VSWR (max)	Power handling capability (max)		Nominal characteristic impedance	Weight (max)	Finish	Figure number
			Average	Peak 1/				
01, 16 6/	GHz DC to 18	(1.05 +.010f):1 2/	Watts .5 3/	Watts 50 3/	Ohms 50	Oz 0.25	4/	1
02, 17 6/	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	5/	1
03, 18 6/	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	4/	2
04, 19 6/	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	5/	2
05	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	4/	3
06	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	5/	3
07	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	4/	4
08	DC to 18	(1.05 +.010f):1 2/	.5 3/	50 3/	50	0.25	5/	4
09, 20 6/	DC to 10	1.10:1	1.0	100	50	0.25	4/	5
10, 21 6/	DC to 10	2.10:1	2.0	200	95	0.25	4/	5
11, 22 6/	DC to 18	1.15:1	1.0	100	50	0.25	4/	1
12, 23 6/	2 to 19	1.30:1	1.0	100	50	0.25	4/	5
13, 24 6/	2 to 19	1.30:1	.5	50	50	0.25	4/	6
14, 25 6/	2 to 19	1.30:1	1.0	100	75	0.40	4/	5
15, 26 6/	4.4 to 5.0	(1.025 +.005f):1 2/	.5	50	50	0.25	4/	1

1/ Peak power is specified with a 1 percent duty cycle.
2/ Where f is the frequency in GHz.
3/ Power input is derated linearly from 100 percent at +25°C to 25 percent at specified maximum ambient operating temperature.
4/ Gold plated in accordance with MIL-G-45204, type II, class 1.
5/ Passivated in accordance with QQ-P-35.
6/ Furnished with safety wire holes (for high vibration areas). Three holes .016 inch (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on coupling nut is optional.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 6.

Electrical characteristics: See table I.

Materials:

Body and connector: Corrosion-resistant steel in accordance with QQ-S-763 or QQ-S-764.

Finish: See table I.

Contact pin and contact socket: Beryllium copper in accordance with ASTM-B-194, 196 or 197.

Finish: Gold plated in accordance with MIL-G-45204, type II, class 2.

Bead chain and lug: Corrosion-resistant steel or plastic.

Weight: See table I.

Ambient temperature range:

Operating:

Dash numbers 01 through 08, 11, 15, 16 through 19, 22 and 26 - -55°C to +125°C.
Dash numbers 09, 10, 12, 13, 14, 20, 21, 23, 24 and 25 - -65°C to +125°C.

Nonoperating (storage): -65°C to +165°C.

Connector tests: The following tests shall be performed after visual and mechanical examination in the qualification inspection.

Force to engage and disengage: Torque 2 inch-pounds, maximum.

Coupling proof torque (for plug male type only): 15 inch-pounds, minimum.

Threaded coupling connector: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Connector durability: 500 cycles at 12 cycles per minute, maximum. After this test, the connector shall meet the engage and disengage requirement.

Vibration: Method 204 of MIL-STD-202, test condition D.

Shock: Method 213 of MIL-STD-202, test condition I.

Barometric pressure: Method 105 of MIL-STD-202, test condition C.

Part or Identifying Number(PIN): M39030/3-(dash number from table I) (S = screened or N = not screened).

NOTES:

1. Revision letters are not used to denote changes due to the extensiveness of the changes.
2. Dummy loads covered by this specification sheet are substitutable for the following manufacturers' part numbers. This information in no way implies that the manufacturer's part is suitable as a substitute for the military PIN.

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Dash number	Manufacturer's code	Manufacturer's part number
01N	97942	580R142H02
01N	16179	20020P
01N	10001	1645650
01N	37695	185629
01N	94987	129129
01N	34228	858B035129
01N	10110	103372
01N	05869	723398-30
01N	16733	691-005
01N	26805	2021-6100-28
01N	82577	928080-1
01N	82577	928080-2
01N	82577	928080-3
01N	82577	928080-4
01N	82577	928080-5
01N	82577	928080-6
01N	96214	534163-1
01N	30003	247AS-C0488-002
01N	94117	292027P1
01N	09408	9022
01N	24602	4110-P
01N	97942	580R142H04
03N	05869	723796-1
03N	16179	20020P/C
03N	16733	691-006
03N	26805	2021-1316-28
03N	96214	534163-1
03N	30003	247AS-C0488-001
03N	09408	9022A
03N	94117	292027P2
03N	05869	713552-1

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Dash number	Manufacturer's code	Manufacturer's part number - Continued.
03N	82577	928080-7
03N	82577	928080-8
03N	82577	928080-9
03N	82577	928080-10
03N	82577	928080-11
03N	82577	928080-12
03N	07397	19-805847-1
03N	24602	4110-PC
04N	05869	713552-3
04N	16733	691-007
04N	26805	2021-5043-28
04N	16179	20020P-9890
05N	05869	723398-31
05N	16733	671-005
05N	26805	2020-6100-28
07N	05869	713552-2
07N	16733	671-006
07N	26805	2020-1316-28
08N	05869	713552-4
08N	16179	20020J-9889
08N	16733	671-007
08N	26805	2020-5103-28
09N	97942	580R142-H01
09N	16179	S20026
09N	95712	680RM
10N	97942	580R142-H03
10N	95712	
10N	93459	
10N	71468	
10N	16179	20020P9450
11N	18203	T180M

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Dash number	Manufacturer's code	Manufacturer's part number - Continued.
12N	97942	580R142-H05
13N	97942	580R142-H06
14N	97942	580R142-H08
15N	93459	M1408

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - AS, OS, SH
Air Force - 19
DLA - ES

Preparing activity:
DLA - ES

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